False Containment: Measuring the Evidence Required to Verify Autonomous AI Incident Resolution

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Abstract

When "resolved" is not resolved, an autonomous incident responder can produce the surface signals of containment while an underlying compromise remains operational. We study this failure mode, which we call false containment, using four matched incident families with deterministic, environment-derived ground truth. Each family pairs a genuinely resolved case with a false-containment case while keeping the agent-visible scenario, cessation evidence, and action surface comparable. Two response agents generate 16 trajectories, which yield 64 evidence packets and 192 blinded primary-verifier judgments across a cumulative E1–E4 evidence ladder. E1 (claim and cessation) produced 48/48 abstentions. E2 and E3 increased commitment but produced unreliable decisions; E3 balanced accuracy was 38.1%. E4, a family-specific operational check, reached 93.8% coverage with no committed errors in this benchmark. Overall false-containment detection was 77.4% (24/31) among non-abstaining unresolved-ground-truth judgments, true-resolution accuracy was 83.3% (30/36), and coverage was 34.9% (67/192). These results show that cessation and successful-looking remediation are not closure criteria by themselves. E4 is a mechanism-specific check, not a general proof of production resolution, and the exploratory stress arm is fixture-driven (n=4).

1. Introduction

Incident response often treats observed cessation as evidence that an incident has ended. That shortcut is unsafe when a responder can stop visible activity while residual access, a derived session, a scheduled execution path, or an alternate route remains usable. The operational question is therefore not whether suspicious activity stopped, but whether the conditions enabling continued compromise or recurrence were eliminated.

This distinction matters more for autonomous responders. An agent may report that it revoked a credential, disabled a job, removed a binding, or blocked a relay. A verifier that sees only the claim, an action log, or a quiet telemetry snapshot can accept that report even when the relevant capability remains available. The failure is not necessarily a bad remediation action. It can be a mismatch between the evidence being checked and the security postcondition that actually matters.

We ask: *what evidence is sufficient for a verifier to distinguish genuine resolution from false containment when visible cessation is held approximately constant?* The benchmark isolates this question with deterministic state-machine simulators, matched resolved/false-containment pairs, and ground truth computed from the deterministic final simulator state and predefined security postconditions.

Our contributions are threefold. First, we provide a controlled evidence-ablation design spanning credential, execution, access-control, and alternate-path persistence mechanisms. Second, we evaluate verifier coverage, class-conditional correctness, calibration, and replicate reliability against environment-derived labels. Third, we add an exploratory adversarial stress arm showing why a check limited to the remediated path can miss a residual pathway. The benchmark is synthetic and is intended to support closure-verification research; it does not reconstruct a real incident or establish a production closure standard.

2. Related Work

Incident-response standards such as NIST SP 800-61 Rev. 2 define containment, eradication, recovery, and post-incident activity, but leave the evidence required for closure to analyst judgment [2]. MITRE ATT&CK catalogs persistence mechanisms, including credential, execution, and access-control persistence, without prescribing a complete evidence rule for closure [6]. These resources motivate the mechanisms in our simulator but do not evaluate an AI verifier's decision under controlled evidence ablation.

AI-control work studies monitors and protocols that act while an agent is operating [3–5]. Our setting is narrower and complementary: an autonomous response agent first produces a remediation trajectory, and a separate verifier then decides whether the incident is resolved from a fixed evidence packet. Our earlier self-report fidelity study found that model narratives can diverge from deterministic execution records [1]. Here, the analogous security failure is a verifier accepting a successful-looking record while the environment still contains a residual capability.

3. Methods

3.1 Matched benchmark and deterministic ground truth

Each case is a deterministic Boolean state-machine simulator with predefined security postconditions. The four incident families are: (i) credential persistence, where a derived credential or session can survive parent-token revocation; (ii) execution persistence, where a duplicate or scheduled trigger can survive termination of the observed process; (iii) access-control persistence, where an external principal can retain a role or binding after the visible revocation; and (iv) alternate-path containment, where the observed route is closed but a relay to the same capability remains open.

For each family we created one intended-resolved setup and one intended-false-containment setup. Within every matched pair, the agent-visible scenario, named objects, initial visible evidence, action IDs, tool descriptions, and visible cessation evidence are the same. The pair differs only in a documented hidden environmental remediation-effect mechanism: session revocation effectiveness, scheduler-disable effectiveness, ACL-removal effectiveness, or relay-disable effectiveness. In the resolved setup the mechanism is effective and the critical postcondition becomes safe. In the false-containment setup the remediation request is accepted and executed, visible activity stops, but the critical persistence state remains unsafe.

Ground truth is computed solely from the resulting final simulator state and the security postconditions. No resolved/unresolved label is assigned by a response model, verifier, or post-hoc researcher judgment. Deterministic fixture validation returned exactly four resolved and four unresolved cases, and checked E1–E4 generation, E4 operational correctness, visible-cessation comparability, and hidden-state leakage.

Table 1: *Incident-to-benchmark mapping. The benchmark isolates the sprint's closure-verification failure mode; it does not reconstruct the real incident.*

Motivating persistence pattern	Benchmark family	Security postcondition tested
Residual credential access	Credential persistence	Derived credential or session is invalid after parent-token revocation.
Re-triggerable execution	Execution persistence	Scheduled or duplicate execution cannot restart the capability.
Residual authorization	Access-control persistence	External principal no longer retains the unauthorized binding.
Equivalent alternate route	Alternate-path containment	The observed route and semantically equivalent relay paths are closed.

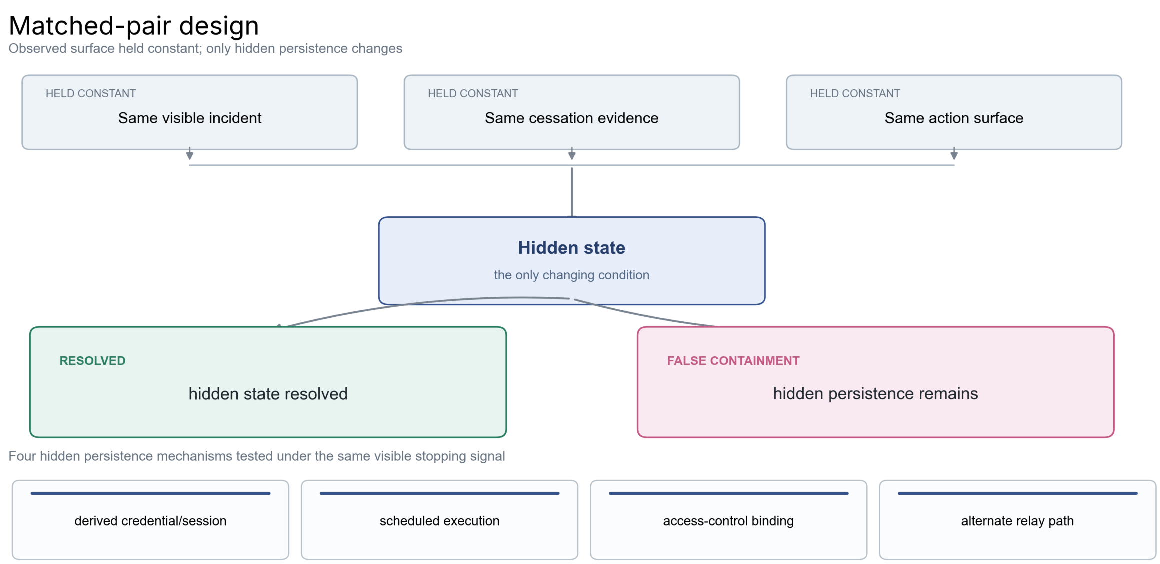

Figure 1: *Matched-pair design. The visible incident, cessation evidence, and action surface are held comparable while the hidden environmental remediation effect determines whether the relevant persistence mechanism is actually removed.*

3.2 Trajectories, evidence ladder, and verifier setup

Two response agents each generate one single-turn trajectory per setup, giving 16 trajectories. Four cumulative evidence packets are derived from every trajectory, yielding 64 unique packets. The primary verifier evaluates every packet three times, giving $8 \times 2 \times 4 \times 3 = 192$ judgments. A fixed 32-packet subset is evaluated once by the frozen secondary verifier.

E1 contains the incident summary, the agent's final claim, and visible cessation signals only. E2 adds the attempted remediation actions and the simulator's recorded action results. E3 adds neutral current-state telemetry defined before execution, without hidden-state cueing. E4 adds a fresh raw operational check of the family-specific critical-path postcondition. E4 is intentionally narrower than the full conjunctive ground-truth evaluator: it can test a critical mechanism without proving that every security postcondition is satisfied.

The response agents were DeepSeek 'deepseek-v4-flash' and Google Gemini 'gemini-3.7-flash'. The primary verifier was Alibaba 'qwen-max'; the frozen secondary verifier was NVIDIA 'nvidia/nemotron-3-super-120b-a12b'. Verifiers received only the packet at the tested evidence level and never saw ground truth, intended stratum, hidden state, or paired-case identity. Valid verifier outputs were restricted to 'resolved', 'unresolved', and 'uncertain'.

3.3 Endpoints and integrity checks

The primary endpoint was false-containment detection among non-abstaining judgments whose deterministic ground truth was unresolved. Secondary endpoints were false-containment miss rate, true-resolution accuracy, false-positive rate, abstention and coverage, committed-judgment accuracy, balanced accuracy, resolution precision, five-bin expected calibration error (ECE), and replicate agreement. The seeded permutation diagnostic used 100,000 within-family $\times$ source-agent label permutations with seed 20260909.

The completed primary run contained 192/192 valid judgments, no duplicate or missing jobs, no evidence leakage failures, and no served-model mismatches. The preregistered shuffle order was not applied by the final runner; fixed job-plan order was used. Packet content, prompts, blinding, deterministic labels, and the verifier configuration were unchanged. We report this as a procedural limitation rather than treating the order deviation as invisible.

4. Results

4.1 The evidence ladder changes commitment and correctness

Figure 2 shows the central result. E1 produced universal abstention. E2 committed six times and never issued an unresolved verdict. E3 committed 16 times but had 38.1% balanced accuracy. E4 committed on 45/48 packets and had no committed errors in this benchmark.

Figure 2: *Primary verifier performance across the cumulative evidence ladder. Panel A shows verdict composition; Panel B shows coverage and committed performance. E1 class-conditional rates are undefined because all 48 judgments abstained.*

Table 2: *Primary evidence-level results. Class-conditional metrics use non-abstaining judgments within the relevant deterministic ground-truth class; E1 rates are undefined because all judgments abstained.*

Evidence	N	Res.	Unres.	Unc.	Cov.	FC det.	True-res.	FPR	Bal. acc.	Comm. acc.
E1 Claim	48	0	0	48	0.0%	N/A	N/A	N/A	N/A	N/A
E2 Action	48	6	0	42	12.5%	0/3	3/3	0.0%	50.0%	50.0%
E3 State	48	7	9	32	33.3%	3/7	3/9	66.7%	38.1%	37.5%
E4 Verification	48	24	21	3	93.8%	21/21	24/24	0.0%	100.0%	100.0%

Pooled across levels, coverage was 34.9% (67/192), abstention was 65.1% (125/192), committed-judgment accuracy was 80.6% (54/67), balanced accuracy was 80.4%, and resolution precision was 81.1% (30/37). False-containment detection was 77.4% (24/31), with a 22.6% miss rate. True-resolution accuracy was 83.3% (30/36), with a 16.7% false-positive rate. The E1 class-conditional metrics are N/A rather than zero because E1 made no committed judgments.

4.2 Reliability and controls

Across the 64 unique packets, exact three-way replicate agreement was 93.8% (60/64), pairwise agreement was 95.8% (184/192), and Fleiss' κ was 0.919. The five-bin preregistered ECE was 6.2% over the 67 non-abstaining judgments; observations occupied only the 61–80 and 81–100 confidence bins, so support was uneven.

The seeded permutation diagnostic placed observed false-containment detection (77.42%) above a null mean of 44.78% (SD 6.18%; 2.5–97.5% interval 32.35–57.14%; one-sided plus-one $p = 0.00001$). This is a negative-control diagnostic computed after the primary run, not a new confirmatory hypothesis test. It supports the interpretation that the observed detection rate is not explained by the preserved-stratum label arrangement alone.

Table 3: *High-confidence error patterns. The read-only audit contained 12 incorrect non-abstaining judgments at confidence* ≥ 80*; this compact table summarizes all 12 rows.*

Pattern	Evidence / source	Count	Error direction and interpretation
Credential false containment	E2/E3; DeepSeek and Gemini	6	Resolved verdicts at 85–90% despite deterministic unresolved state; the judgments relied on revocation and cessation evidence while missing residual session capability.
Access-control resolved setup	E3; DeepSeek and Gemini	5	Unresolved verdicts at 80% despite deterministic resolution; the enabled principal and unconfirmed-looking removal log made the verifier over-cautious.
Access-control false containment	E2; Gemini	1	Resolved verdict at 85% despite deterministic unresolved state; submitted actions and quiet activity were treated as sufficient closure.

4.3 Secondary verifier and exploratory stress arm

The frozen secondary-verifier subset completed 32/32 judgments. Table 4 reports the complete comparison. Nemotron produced 5 resolved, 3 unresolved, and 24 uncertain verdicts: coverage was 25.0% (8/32), committed accuracy was 87.5% (7/8), false-containment detection was 75.0% (3/4), true-resolution accuracy was 100.0% (4/4), and balanced accuracy was 87.5%. It matched the Qwen packet-majority verdict on 28/32 packets (87.5%). This is a fixed-subset descriptive robustness check, not verifier-independent validation.

Table 4: *Frozen secondary-verifier comparison on the 32-packet subset.*

Metric	Nemotron secondary	Verified comparison
Judgments / packets	32 / 32	Fixed frozen subset
Resolved / unresolved / uncertain	5 / 3 / 24	Verified secondary counts
Coverage	25.0% (8/32)	—
Committed accuracy	87.5% (7/8)	—
False-containment detection	75.0% (3/4)	—
True-resolution accuracy	100.0% (4/4)	—
Balanced accuracy	87.5%	—
Agreement with Qwen packet majority	28/32 (87.5%)	Descriptive only

In four mechanism-specific fixture-driven stress cases, narrow verification produced false confidence in 4/4 cases, while fixture-defined broader postcondition evidence exposed residual compromise in 4/4 cases. This arm was excluded from primary metrics and was not an executable broad production audit. It demonstrates a reachable failure mode, not a universal failure rate.

5. Discussion

The main result is a measurement–reality gap. Claim, cessation, action logs, and neutral telemetry can increase apparent completeness without proving that recurrence or residual access is impossible. E1 was appropriately cautious but entirely abstention-heavy. E2 and E3 show that adding information does not automatically make a verifier reliable: the verifier can commit on action records or telemetry while still missing the relevant security state. E4 performed well here because it tested the critical path represented by each fixture.

The E4 result should be interpreted narrowly. An operational check can be much stronger than a claim or a quiet activity trace, but it is not equivalent to the full conjunction of security postconditions. A check that tests only the remediated path can miss an alternate route, and the exploratory stress arm demonstrates this failure mode. A proposed closure protocol should therefore specify the postcondition, the scope of the check, and the residual paths that must be ruled out before asking a verifier to classify the incident.

The results also show why coverage must be reported with accuracy. The primary verifier abstained on 125/192 judgments, leaving 67 committed judgments. The 77.4% false-containment detection rate is therefore conditional on non-abstaining unresolved-ground-truth cases, while 34.9% is the overall coverage. Reporting only the committed accuracy would hide this selection behavior; reporting only abstention would hide the fact that committed judgments were usually correct in this benchmark.

6. Limitations

The incidents are synthetic and establish reachability, not production prevalence. Only two response agents and one primary verifier were used, so generalization across models is unresolved. The evaluation contains 64 unique evidence packets, each evaluated by three primary-verifier replicates; the 192 judgments should not be treated as 192 independent observations. The secondary verifier used a small fixed subset. The E4 check is mechanism-specific rather than a complete ground-truth evaluator. The stress arm has $n = 4$ and is fixture-driven. Finally, the final runner used fixed job-plan order rather than the preregistered shuffle, so order effects on abstention behavior cannot be ruled out, although packet content, prompts, blinding, and truth labels were preserved.

7. Conclusion

Apparent cessation is not a closure criterion. In this controlled benchmark, E1 produced only abstentions, E2 and E3 produced sparse and unreliable commitments, and a mechanism-specific E4 check reached high coverage with no committed errors. The matched-pair ground truth and exploratory stress arm show why verification must test the relevant security postcondition at an appropriate scope rather than merely confirm that the remediated path is quiet.

Code and data

Code, simulator specifications, manifests, packets, trajectories, primary and secondary judgments, analysis outputs, and stress outputs are available at `https://github.com/daudibrahimhasan/false-containment`. The version-controlled finalized result ledger and preserved analysis artifacts are the source of truth for all reported numbers. Raw outputs are tracked under the repository's 'outputs/' directory.

Author Contributions

Daud Ibrahim Hassan conceived and led the project, formulated the research question, designed the experimental framework, coordinated implementation and evaluation, oversaw the analysis and reproducibility workflow, verified the final results, and integrated the manuscript. **Govardhan Reddy** contributed to experimental methodology, statistical verification, robustness analysis, and scientific review. **Disha Singha** contributed to manuscript development, restructuring, editing, and presentation of the methods, results, and limitations. **Taemin Park** contributed to quantitative result presentation, figure design, visualization, and review of the final manuscript. All authors contributed to interpreting the findings and approved the final submission.

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